

A Machine-Checked Companion to Goertzel’s Spencer–Brown/Kauffman Route to the Four-Color Theorem: Keystone, Pillars, and the Composition Ladder

The MeTTapedia formalization effort

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Abstract

This is a living formalization companion to Ben Goertzel’s manuscript *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem* (v23), which organizes a candidate human-readable proof of the Four-Color Theorem (4CT) around three interacting pillars: (A) an annular difference-*spanning* statement for a zero-boundary space $W_0(H)$ (Theorem 4.9); (B) a CAP5-free normal form with radius-1 collar locality; and (C) an *executability* layer that, refusing a false additivity of Kempe switches, reduces between-region Kempe connectivity to a single *Local Kempe Reachability with fixed input* (LKR_{in}) property for one canonical three-cell collar gadget τ (Lemma 8.14), composes it along the collar chain (Lemmas 8.15, 8.18, Cor. 8.19), and assembles the wide annulus collar-by-collar (§9.2). We record what the MeTTapedia formalization has machine-checked of this route, with an honest ledger separating what holds outright, what holds modulo one named remainder, and what is open, and we foreground the places where the formalization *refutes* a step rather than discharging it.

Machine-checked outright (no `sorry`; axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$): the keystone LKR_{in} audit of the canonical gadget (Lemma 8.14) and its two-color *transparency* (Lemma 8.15); the *need-based repair* of Pillar A — the manuscript’s spanning *equality* is refuted in exactly the direction the synthesis never uses (a proper-subspace obstruction on closed-walk shells, with four uniform repairs refuted), while the *inclusion* the route actually consumes is proved from the manuscript’s own peeling witnesses, Kempe-switch steps are proved to preserve the detector subspace, and route discrepancy classes are proved to lie in it; and the closed-collar *S2 bridge* (§9.2 Step 3), certifying that all length-2/3 re-glued collar fibers are Kempe-connected. The Gate-2 composition (Lemmas 8.18/8.19) is *reduced* — the manuscript’s own pointwise Step-1 preparation is refuted and replaced by a frontier-automaton / fibration architecture — to a single named structural remainder.

The formalization also produced a decisive negative. §9.2 Step 4 — the manuscript’s reinterpretation of collar Kempe words as words of the ambient annulus — is refuted by an explicit two-layer counterexample: a certified, collar-input-avoiding switch whose ambient component escapes the inner boundary, re-enters, and changes the collar restriction. This is a defect in the composition *seam* as written, not a refutation of 4CT; its salvage (a normal-form confinement argument, or a face-collar redesign) is open. We make no claim about the Four-Color Theorem — which is of course a theorem, via the Appel–Haken and Robertson–Sanders–Seymour–Thomas computer-assisted proofs. The human-readable v23 route remains *open*, now with its foundational layer machine-checked and its most dangerous assembly seam pinned down as broken.

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1 The problem and Ben Goertzel’s three-pillar route

1.1 Tait’s reduction and the minimal-counterexample target

The Four-Color Theorem asserts that every plane triangulation G admits a proper vertex 4-coloring. Passing to the cubic dual $K = G^*$, Tait’s classical equivalence (v23, Thm. 2.3) restates 4CT as: every bridgeless planar cubic graph admits a *proper 3-edge-coloring*, i.e. a map $\chi : E(K) \rightarrow \{r, b, p\}$ assigning distinct colors to the three edges at each vertex. The manuscript encodes the three colors as the three nonzero elements of $\mathbb{G} = \mathbb{F}_2^2$, with $r = (1, 0)$, $b = (0, 1)$, $p = (1, 1)$, so that “properness at a vertex” is exactly $\chi(e_1) + \chi(e_2) + \chi(e_3) = r + b + p = 0$ in $\mathbb{G}$, and “integrating” an edge-coloring across the planar dual becomes a path-sum in $\mathbb{F}_2^2$.

Following Spencer–Brown and Kauffman, the manuscript reformulates colorings as plane *formations* (red/blue Jordan curves with purple overlaps), defines Kempe switches as the standard two-color interchange, and proves the *Parity Lemma* (planar invariance of the mod-2 count of two-color circuits). The role of §3 is to reduce 4CT, via Kauffman’s parity/primalty argument (Theorem 3.1), to the nonexistence of a minimal counterexample containing a *between-region annulus* H whose boundary-compatible colorings are *not* connected by Kempe moves supported inside the annulus. Thus the entire analytic burden is shifted to a *between-region Kempe connectivity* statement for an annulus $H \subseteq K$ with outer and inner boundary cycles Γ_∞, Γ_0 .

1.2 The three pillars

The manuscript’s conceptual overview (v23, §1.1) states the argument as three interacting pillars.

Pillar A: annular difference spanning (the linear “accounting” layer). Working in a between-region annulus H , one encodes boundary colorings and their differences as vectors in an $\mathbb{F}_2$ -space. Let $W(H)$ be the coordinatewise cycle space of $\mathbb{G}$ -chains on $E(H)$ satisfying the Kirchhoff/even-degree constraints at interior vertices, and let

$$W_0(H) := \{x \in W(H) : x(e) = 0 \text{ for all boundary edges } e \in B_1 \cup B_2\}$$

be the *zero-boundary* subspace. For a fixed proper coloring C_0 of H with Kempe closure $\text{Cl}(C_0)$, the manuscript defines a family $\text{KC}(C_0)$ of *Kempe-closure (polarized face) generators* $X_f^{ab}(C)$ supported in depth-1 collars (v23, Def. 4.8), and proves:

Theorem 1 (Spanning theorem for $W_0(H)$, v23 Theorem 4.9). *Let H be an annular subgraph of a bridgeless planar cubic graph, with boundary cycles Γ_0, Γ_∞ . Fix any proper 3-edge-coloring C_0 of H and let $\mathcal{G} := \text{KC}(C_0)$. Then $W_0(H) \subseteq \text{span}_{\mathbb{F}_2}(\mathcal{G})$; equivalently, if $z \in W_0(H)$ satisfies $z \cdot g = 0$ for every $g \in \mathcal{G}$, then $z = 0$.*

The manuscript proves Theorem 1 in strong-dual annihilator form: it shows $U^\perp \cap W_0(H) = \{0\}$ for $U = \text{span}_{\mathbb{F}_2}(\mathcal{G})$, by a planar orthogonality peeling along a dual spanning forest, after a “purification” step (Lemma 4.4) that removes the polarization of the face generators.

Pillar B: CAP5-free geometry and radius-1 locality (the geometric locality layer). Assuming a minimal counterexample, the manuscript eliminates the CAP5 separating obstruction and works in a CAP5-free normal form (Corollary 6.9). A tube-forcing argument then reduces every “escape” of an anchored Kempe chain to a finite three-cell gate enumeration, yielding radius-1 confinement of anchored Kempe cycles together with a cap-aware disjointness/commutativity package (Lemmas 7.8, 7.9, 7.10). This is what licenses peeling the wide annulus into radius-1 collars.

Pillar C: executability without false additivity (the algebra-to-geometry bridge). The decomposition $\Delta = \sum v_{f,g}$ provided by Pillar A is a statement in $W_0(H)$; it does *not* by itself imply that one may *execute* the corresponding mixed color-pair Kempe switches as if they formed an abelian sum. The manuscript is explicit that cross-pair interference is real: an (a, b) -switch can destroy the (b, c) -structure a later switch needs (v23, §8.1), and Kempe switches using different color pairs on shared support need not commute (v23, §8.3). To execute the decomposition honestly, the manuscript moves to an operator model: proper colorings form an orthonormal basis of a Penrose–Kauffman state space, and each Kempe switch is an involutive linear operator (generally noncommuting with switches of other color pairs). Using radius-1 locality, between-region Kempe connectivity is reduced to a single local property:

Definition 1 (Local Kempe reachability with fixed input, v23 Def. 8.12). Let X be a planar cubic tangle with ordered input stubs S_{in} and output stubs S_{out} , and fix an input coloring c . Let $\Omega_X(c)$ be the set of proper 3-edge-colorings of X extending c on the input stubs. The *Kempe graph* $K_X(c)$ has vertex set $\Omega_X(c)$, with an edge whenever two colorings differ by a single Kempe switch supported on a two-color component touching no input-stub edge. We say X satisfies LKR_{in} if $K_X(c)$ is connected for every c for which $\Omega_X(c) \neq \emptyset$.

The chain of Pillar C is then:

- **Lemma 8.14:** the canonical three-cell collar gadget τ satisfies LKR_{in} (and so does its mirror);

- **Lemma 8.15:** two-color *transparency* of τ — every two-color component is a boundary-to-boundary path with no internal cycle;
- **Lemma 8.18 / Corollary 8.19:** LKR_{in} *composes* across a transparent interface, so an entire cut-open radius-1 collar tangle $\hat{A} = G_m \circ \cdots \circ G_1$ satisfies LKR_{in} ;
- **§9.2 assembly:** peel the wide annulus H into nested radius-1 collars and use the collar LKR_{in} property to rewrite the current coloring to the target collar-by-collar from outside to inside, the cap-aware locality lemma (7.9) ensuring no switched component reaches back to change a fixed boundary.

1.3 Why Lemma 8.14 is the keystone

The manuscript states that its proof is human-readable and “the only finite verification required is a small by-hand audit of a single three-cell gadget” (v23, Abstract and §1). In the three-pillar architecture, every other step is either an explicit local lemma or a structural reduction; the one place where a finite, exhaustive check is genuinely unavoidable is the LKR_{in} property of the canonical gadget τ . Formalizing Lemma 8.14 therefore certifies exactly the finite core that the manuscript isolates as load-bearing. The remainder of this companion describes that formalization (§2), the resolution of the Pillar A spanning question into a need-based repair (§3), the Pillar C composition ladder — what is proved, what is reduced to one pin, and where a decisive counterexample was found (§4) — and a precise status ledger (§5). Superseded routes are recorded in appendices.

2 The Pillar-C keystone, machine-checked: Lemma 8.14 and transparency 8.15

We restate the manuscript’s keystone in its own notation and then describe the Lean encoding.

Lemma 1 (LKR_{in} for the canonical three-cell gadget, v23 Lemma 8.14). *Let τ be the canonical three-cell collar gadget. It is a planar cubic tangle with internal vertices $F_0, \dots, F_5$ and boundary stubs $B_0, \dots, B_7$, whose internal edges form the tree*

$$F_1 - F_0 - F_2 - F_4 - F_5, \quad \text{with a leaf } F_3 \text{ attached to } F_2,$$

and whose boundary stubs are attached by $F_0B_0, F_1B_1, F_1B_2, F_3B_3, F_3B_4, F_5B_5, F_5B_6, F_4B_7$. Take input stubs $\{B_0, B_1, B_2, B_3\}$ and output stubs $\{B_4, B_5, B_6, B_7\}$. Then τ satisfies LKR_{in} . The same holds for the mirror gadget.

The manuscript proves Lemma 1 by cutting τ along the internal interface edge $I = F_2F_4$ into $\tau = \sigma_{\text{R}} \circ \sigma_{\text{L}}$ and doing a small case analysis on the two pieces. Our formalization instead audits the *full* gadget directly, by exhibiting the proper-coloring state space and an explicit Kempe-path certificate connecting every fixed-input fiber; this is a strictly more direct realization of the same LKR_{in} claim, and it does not rely on the $\sigma_{\text{L}}, \sigma_{\text{R}}$ split or on the composition Lemma 8.18.

2.1 The canonical gadget and its 192-state coloring space

The Lean module fixes the colors and the gadget incidence exactly as in Lemma 1: the three nonzero colors are an inductive type, and the gadget’s edges (five internal edges plus the eight boundary stubs) and internal vertices are inductive types with an explicit incidence map.

```

inductive LColor | r | b | p
  deriving DecidableEq, BEq, Repr, Inhabited

inductive TauEdge
  | F1F0 | F0F2 | F2F3 | F2F4 | F4F5
  | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7
  deriving DecidableEq, BEq, Repr, Inhabited

def tauInputEdges : List TauEdge := [TauEdge.B0, TauEdge.B1, TauEdge.B2, TauEdge.B3]

```

A proper 3-edge-coloring of τ is represented by the finite choices left after properness is imposed at every cubic vertex: an ordered color triple at the central vertex F_2 , one binary choice for the forced edge F_1F_0 , and one ordered-pair (swap) bit at each remaining boundary fork — $6 \times 2^5 = 192$ states. That this enumeration is exactly the proper-coloring state space is a checked fact (192 entries, no duplicates, proper at every vertex).

2.2 The Kempe audit and the machine-checked theorems

A Kempe step switches a two-color component *disjoint from the input stubs* (Definition 1). The LKR_{in} property is witnessed by an explicit indexed-path certificate: for each of the 192 states it records the representative of that state’s fixed-input fiber and a path of at most two input-disjoint Kempe switches reaching it, plus a coherence audit that any two states with the same input edges share a representative (so fibers are well defined). The three finite audits each reduce to **true** by the kernel’s decision procedure; the only axiom dependency is **propext**.

```

theorem tauStateSpaceAudit_ok : tauStateSpaceAudit = true := by decide
theorem lemma814PathCertificate_rows_ok :
  validIndexedRowsFrom 0 lemma814PathCertificate = true := by decide
theorem representativeCoherenceAudit_ok : representativeCoherenceAudit = true := by
  decide

/-- Lemma 8.14, closed by the finite state-space and indexed path certificate. -/
theorem lemma814_tau_lkrIn_audit : lemma814_tau_lkrIn_finiteCheck = true := by
  simp [lemma814_tau_lkrIn_finiteCheck, tauStateSpaceAudit_ok, tauLKRInAudit,
    lemma814PathCertificateAudit_ok]

```

The mirror clause holds definitionally (reflection changes only the planar order of an isomorphic incidence tree). Thus Lemma 8.14 — the “only finite verification 4CT requires” — is machine-checked in module `GoertzelLemma814`.

Remark 1 (Transparency 8.15, now machine-checked). *The same module formalizes Lemma 8.15 (two-color transparency): for each of the 192 colorings, every nonempty two-color component is a boundary-to-boundary tree path (**transparentComponent**), aggregated as **tauTreeTransparencyAudit**. That audit is now proved — **lemma815_tau_tree_transparency_audit** reduces to **true** by a range decomposition of the finite check, axiom-clean. Transparency is what licenses the composition step (§4); with Lemma 8.14 it is the sealed local core of Pillar C.*

3 Pillar A: a proper-subspace obstruction, and its need-based repair

Pillar A (the spanning Theorem 1) is where the formalization first found — and then routed around — a real gap. We give both halves, because together they are a clean example of what a companion of this kind is for.

3.1 The obstruction: the spanning *equality* over-reaches

On every *closed-walk exact shell* — a concrete annular configuration carrying the boundary-source and interior-endpoint witness data that Theorem 1 is meant to cover — the boundary-zero target that the Kempe-closure detector actually pins down is a *strict* subspace of the full boundary-zero chain space.

```
-- On every closed-walk exact shell, the Theorem-4.9 boundary-zero target is a
proper subspace of the full selected-boundary-zero chain space. -/
theorem
  theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule_of_closedWalkExactShell

  [Fintype G.edgeSet] {emb : PlaneEmbeddingWithBoundary G}
  (shell : ClosedWalkExactShell emb) :
  theorem49BoundaryZeroKirchhoffSubspace emb
    (selectedBoundaryInteriorEdgeEndpointVertices emb) <
    planarBoundaryZeroSubmodule emb
```

The strict inequality < is the load-bearing content: a proved *proper*-subspace inclusion, not a heuristic. The structural reason (every closed-walk shell forces two distinct interior-support edges onto one face boundary) refutes the natural one-collar repairs, and a companion audit shows the four proposed *uniform geometric oracles* that would each close Theorem 1’s synthesis are all false:

```
theorem not_interiorDualPeelOracle      : ¬ InteriorDualPeelOracle
theorem not_wellFoundedPeelOracle       : ¬ WellFoundedPeelOracle
theorem not_previousBoundaryGeometryOracle : ¬ PreviousBoundaryGeometryOracle
theorem not_residualBoundaryGeometryOracle : ¬ ResidualBoundaryGeometryOracle
```

So the spanning claim, in the form $W_0(H) = \text{span}_{\mathbb{F}_2}(\mathcal{G})$, does not hold as stated, and the obvious uniform routes to bridge the strict gap do not exist.

3.2 The repair: the route never needs the full space

The resolution is that the synthesis does not *use* the full boundary-zero space; it uses only the discrepancy classes of boundary-compatible coloring pairs, and those lie in the detector subspace the manuscript’s own peeling witnesses span. Formally, the requirement actually consumed downstream is *need-based*: it suffices that (1) the detector subspace is contained in the projected Kempe-closure generator span, (2) each execution step preserves that subspace, and (3) every route-used discrepancy class lies in it. All three are machine-checked in the route’s natural cubic setting (module PillarASmallerSpace):

```
-- (3) discrepancy classes lie in the detector subspace (cubic setting):
theorem routeUsedDiscrepancyClassesLieInDetectorSubspace_of_hasCubicIncidentEdgeTriples
  ...
```

```

-- (2) a Kempe-switch execution step preserves detector membership:
theorem routeKempeSwitchExecutionStep_preserves_detectorSubspace ...
-- assembled sufficiency (inclusion form), discharging the requirement:
theorem
  needBasedPillarARequirement_of_detector_le_span_of_stepPreserves_of_routeDiscrepancies
  ...

```

Proposition (3) has a two-idea proof one can state aloud: a Tait coloring is a Klein-four flow at each cubic vertex, so two colorings agreeing on the selected boundary differ by a *boundary-zero flow*, which is exactly what the Kirchhoff detector certifies. The manuscript’s Theorem 4.9 *equality* was mis-stated in the direction the route never uses; its *inclusion* — detector $\subseteq$ span, proved from the manuscript’s own peeling witnesses — is the direction it does use. What remains a genuine per-embedding obligation is the *supply* of a peeling witness for the synthesis’s actual embedding class; that is a composition-stage task, not a spanning gap.

4 Pillar C: the composition ladder

Between the local keystone (§2) and 4CT, v23’s own argument climbs a ladder: single-gadget LKR_{in} → transparent composition to a whole collar chain (Gate-2) → closed re-glued collars (§9.2 Step 3) → execution in the ambient annulus (§9.2 Step 4) → the wide-annulus peel. This companion has now walked a substantial part of that ladder, with one step proved-modulo-a-pin and one step *refuted*.

4.1 Gate-2: transparent composition, reduced to one pin

Corollary 8.19 lifts LKR_{in} from the single gadget to a whole cut-open radius-1 collar chain across a transparent interface. The manuscript’s own Step-1 proof of Lemma 8.18 (a pointwise preparation via Lemma 8.16) does *not* go through — the formalization refutes it directly:

```

theorem lemma818_pointwise_lift_preparation_obstruction : ... -- axiom-clean

```

In its place the development builds a frontier-automaton / fibration architecture: a finite frontier state over the chain transcript, generated parent-row certificates, and a fibered-connectivity decomposition (base connected + fibers connected + lifts). The executable chain test `chainLKRInAudit` passes on all words up to length seven (real-fiber labs, all input fibers connected, zero failures), and Gate-2 is reduced to a *single* named structural remainder,

```

concreteChainFiberAppendQuotientFibrationParentRowsNonSingletonPrefixStructuralClosed

```

carried as a hypothesis through the semantic bridge. (A separate refuted route, `not_concreteChainFiberAppendPrefixFibrationClosed`, records a shortcut that provably fails and is kept dead.)

4.2 S2: the closed-collar bridge, proved

Gate-2 certifies LKR_{in} with the *first* gadget’s input stubs fixed; §9.2 Step 3 needs the *cyclically re-glued* collar with all outer-side stubs fixed — different fibers, no bridging lemma in the manuscript. On the finite length-2/3 tau/mirror class this is now *proved*: after a lab found all 972 closed-collar fibers connected-or-empty (216 nonempty, 756 empty, zero disconnected), the target was discharged with zero pins.

```

theorem closedCollarS2BridgeTarget_proved : closedCollarS2BridgeTarget
theorem section92S2BridgeLemmaTarget_proved : section92S2BridgeLemmaTarget

```

The route survey had rated S2 the sharpest mismatch in the manuscript — a statement never written down in v23 and not implied by Gate-2 — so proving it on the finite class is a real advance.

4.3 S4: execution in the ambient annulus, refuted

§9.2 Step 4 reinterprets a collar Kempe word as a word of the *ambient* annulus, and confines the switched components with a face-only locality lemma (7.9). The formalization tested this on the smallest honest geometry — a collar plus one deeper radial ring sharing the interface — and found a decisive counterexample.

```

/-- A collar-certified, collar-input-avoiding Kempe switch whose ambient
two-layer component leaves through the inner side, hits the fixed outer input,
and does NOT restrict to the certified collar move. -/
theorem section92Step4TwoLayerCounterexampleClass :
  Section92Step4TwoLayerCounterexampleClass      -- axioms [proptext]

```

Concretely (outer word $\tau\tau$, inner word τ , input key r, r, b, r): a switch that is input-avoiding *in the collar alone* balloons, in the two-layer object, into a component that crosses the shared inner boundary into the deeper ring, returns, meets the fixed outer input edge, and lands the collar on a *different* coloring than the certified move. This refutes the reinterpretation as a regime-free, purely-local principle: the ambient switch is a *different* move on the collar than the one Gate-2/S2 certified.

We are precise about what this does and does not say. It is a defect in the manuscript’s assembly *seam* as argued (Step 4 invokes 7.9, a lemma proved for single-face collars, as if it were a global confinement), not a refutation of 4CT.

The cheapest repair has also been tested, and *refuted*. Strengthening the certification so that switches avoid the entire outer interface — both input *and* output stubs, not only the inputs — does eliminate the escape (a two-layer companion run finds zero escapes under full-interface avoidance), but at a fatal cost: on the same finite class, two colorings with the same fixed input are connected *only* by a switch whose component touches the output stubs, so forbidding output touches disconnects the fibers (module `GoertzelLemma818TwoSidedAvoidance`: 432 open and 216 closed nonempty fibers disconnect). The escape channel is therefore *load-bearing for connectivity*: the very moves that let a component escape the collar are the moves the fibers need to remain connected. Confinement and connectivity are in direct tension, so any repair must confine the escaping components *without* severing collar connectivity — a needle the simplest fix cannot thread.

What remains is a genuinely regime-level salvage: a proof that the minimal-counterexample *regime* (CAP5-free normal form) excludes the disconnecting/escaping configurations, or a §9 re-design that only ever peels *face* collars where 7.9 applies as stated. Neither is established; both need planar-normal-form infrastructure the development does not yet have; and the abstract gadget lab cannot decide them, since the tension may or may not dissolve in the actual normal form. As of this writing the seam is open, with both the escape and the failure of the cheapest repair pinned down as checked theorems.

4.4 Sharpening the open salvage: a winding invariant and a cyclic 2-cut reduction

Since the escape was pinned down, the formalization has *sharpened* — though not closed — the open salvage, by asking a Spencer–Brown/Kauffman-native question: which two-color components can escape? On the closed annular collar a two-color component carries a well-defined mod-2 *winding class* around the annulus generator — the parity of its crossing count with a fixed radial cut — a Parity-Lemma-style invariant, well defined per component and tracked across Kempe switches. In this language the escape is exactly a *winding* move: the components that leave through the inner side and return are those of nonzero winding class. The precise form of the S4 freedom is then a homological statement, machine-checked on the witness class: *winding is not boundary-determined* — a fixed-boundary fiber can carry components of different total winding class, so the escaping (winding-1) moves are not forced by the boundary data alone.

This localizes the difficulty to one geometric question: can a winding-freedom escape be *realized on a genuinely simple planar collar*? The formalization reduces it to a classical connectivity fact. Any simple planar realization that preserves the winding profile must replace the escape’s parallel endpoint bundle by a simple two-pole side; a tree-rank (handshaking) count — $2|\text{side edges}| = 3|\text{internal}| + 4$ forces $|\text{internal}| + 1 < |\text{side edges}|$, hence a cycle — turns that side into a *cyclic 2-cut* of the collar. A collar admitting no realized cyclic 2-cut therefore admits no escape realization:

```
-- every simple planar winding-escape realization forces a cyclic two-cut:
theorem ...Escape_forces_cyclicTwoCut_allSizes : ...
-- so a collar forbidding cyclic two-cuts (e.g. cyclically 5-edge-connected)
-- admits no escape realization:
theorem ...Escape_not_simplyRealizable_allSizes
  (h : ...ForbidsCyclicTwoCut G) : ¬ ...SimplePlanarEscapeRealization G
```

We are precise about what this buys. It is a *reduction of the open salvage to one named residual*, not a repair: the theorem is conditional on the collar forbidding cyclic 2-cuts, and the current realization datum carries the separating-pair structure as an obligation rather than deriving it from the collar’s embedding. What it does is convert the vague “the minimal-counterexample regime should exclude the escape” into a concrete checkable target — *the escape collar carries no cyclic 2-cut*, i.e. it is cyclically sufficiently edge-connected, which is exactly the kind of property the minimal-counterexample normal form is expected to supply. And this residual is not an unformalizable abstraction: the development already proves cyclic-five-edge-connectivity for concrete graphs (and refutes it for others), so discharging or refuting it *for the escape collar* is a reachable next question — the honest current frontier of the S4 salvage.

5 Status and scope: machine-checked, reduced, and open

We state the scope exactly. The full route is a ladder $S_0, \dots, S_9$ (Gate-2, then v23 §9 and §3); the table records each step’s manuscript reference and current Lean status.

Step	Content	v23 ref	Lean status
—	single-gadget LKR_{in} (keystone)	Lem. 8.14	proved
—	two-color transparency	Lem. 8.15	proved
—	Pillar A, need-based	Thm. 4.9 (repaired)	proved (inclusion); equality <i>refuted</i>
S0	Gate-2 chain LKR_{in}	Lem. 8.18/8.19	one named remainder pin
S2	closed-collar bridge (finite class)	§9.2 Step 3	proved (972 fibers, 0 pins)
S4	ambient execution	§9.2 Step 4	refuted (two-layer witness)
S1	collar $\Rightarrow \tau$ /mirror chain	Lem. 8.5–8.6	open (structural)
S3	annulus peeling well-definedness	§9.2 Step 2	open (structural)
S5	interface to Kauffman Thm. 3.1	§9.2 / §3	open (statement-level)
S6	Kauffman parity/primality internals	§3.2–3.3	open (structural)
S7	minimal-counterexample normal form	Lem. 5.2–5.3	open (structural)
S8	CAP5 reducibility	§6	partial
S9	minimality wrap (+ Tait)	Def. 5.1 / Thm. 2.3	open (structural)

Machine-checked outright. The keystone Lemma 8.14 and transparency Lemma 8.15; the need-based repair of Pillar A (the equality refuted on shells with four uniform repairs refuted; the inclusion, step-preservation, and discrepancy-membership proved); and the closed-collar S2 bridge on the finite length-2/3 class. All with no `sorry` and axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

Reduced to one pin. Gate-2 (Lemmas 8.18/8.19): the manuscript’s pointwise Step-1 preparation is refuted and replaced by the frontier-automaton / fibration architecture, whose real-fiber labs pass to length seven; the whole composition rests on a single named structural remainder.

Refuted (a confirmed defect, not a claim). §9.2 Step 4’s collar-word-to-ambient-word reinterpretation, by an explicit two-layer escape counterexample. This lowers the odds that the v23 route completes *as written*; its salvage is open.

Open (the non-finite heart). Steps S1, S3, S5–S9: the realization of abstract chains as actual collars, the wide-annulus peeling, the interface to Kauffman’s reduction, and the Kauffman/normal-form/CAP5 internals. Their common cost is one shared planar-embedding/duality/Jordan infrastructure the development does not yet have; most carry a known proof shape, so their risk is formalization cost rather than mathematical doubt — with the exception of the S4 salvage, which is genuinely open.

What is not claimed. We do *not* claim the Four-Color Theorem, and we do not claim that v23 proves it. The 4CT is a theorem via the computer-assisted proofs; what is at issue is solely the viability of the v23 human-readable route, which remains *open* — its foundational layer now machine-checked, its most dangerous assembly seam pinned down as broken.

Provenance

The manuscript discussed is Ben Goertzel, *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem*, version v23. The formalization is work of the MeTTa-pedia formalization effort. The keystone and transparency are module `GoertzelLemma814`; the Pillar-A obstruction and its need-based repair are `FrontierGeometric`, `GoalAudit`, and `PillarASmallerSpace`; the Gate-2 architecture is `GoertzelLemma818SemanticBridge` and the

frontier-automaton modules; the S2 bridge is `GoertzelLemma818ClosedCollarS2Bridge`; the S4 counterexample and the refuted two-sided-avoidance repair are `GoertzelLemma818TwoLayerEscape` and `GoertzelLemma818TwoSidedAvoidance`. Lean listings are ASCII transliterations of the source declarations (the sources are normative); every headline was checked with `#print axioms`. Per-declaration source links are omitted pending publication of the development tree.

A A graveyard of superseded sub-routes

Each entry is a route or reading that was tried and set aside; several are themselves machine-checked theorems that a stated step *fails*.

The spanning *equality* $W_0(H) = \text{span}(\mathcal{G})$. Refuted on every closed-walk exact shell (§3); superseded by the need-based inclusion, which is what the synthesis actually consumes.

The four uniform geometric peel oracles. Each would have closed Theorem 4.9’s synthesis uniformly; all four are machine-checked false. The route uses the manuscript’s own per-embedding peeling witnesses instead.

The pointwise Step-1 preparation of Lemma 8.18. The manuscript’s own pointwise lift (via Lemma 8.16) is refuted (`lemma818_pointwise_lift_preparation_obstruction`); replaced by the frontier-automaton / fibration composition.

The prefix-fibration shortcut for Gate-2. A tempting closed-form shortcut, machine-checked to fail (`not_concreteChainFiberAppendPrefixFibrationClosed`); kept dead.

The false “additivity” of Kempe switches. An $\mathbb{F}_2$ -additive execution of the Pillar-A decomposition $\Delta = \sum v_{f,g}$. This is *not* the manuscript’s route — Pillar C is titled “executability without false additivity,” and v23 §8.1–8.3 show mixed color-pair switches on shared support do not commute. Recorded only to set aside; the faithful route is the operator/ LKR_{in} execution of §2.

An $\mathbb{F}_2$ -additivity / evader reconstruction (disavowed). An early exploratory reconstruction outside the manuscript’s route, which v23 explicitly disavows (additivity invalid for mixed color-pairs; Rmk. 8.10). Not part of this companion’s formalized content.